Object-Conditioned Strategy-Level Parameter Configuration for Haptic Teleoperation of Heterogeneous Objects

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Abstract

Fixed teleoperation parameters may not provide suitable interaction behavior across objects with different fragility, geometry, and grasping requirements. Existing object-conditioned approaches mainly adjust slave-side impedance, whereas joint configuration of slave-side impedance, master-side haptic feedback, and gripper execution parameters has received limited attention. To address this issue, this paper proposes an object-conditioned strategy-level parameter-configuration framework for haptic teleoperation of heterogeneous objects. Each detected object class is mapped to one of three manipulation strategies, which jointly specify the parameter settings of the three channels. Low-rate visual perception is integrated asynchronously with a nominal 200-Hz supervisory loop. The first mapped detection satisfying the confidence threshold triggers a one-shot parameter update, and the selected strategy remains locked for the remainder of the trial. The framework was evaluated in a human-in-the-loop experiment involving three operators, six objects, five control modes, and 135 grasp-and-transfer trials arranged in 27 matched blocks. Compared with impedance-only reconfiguration, the complete joint configuration reduced mean task completion time by 1.79 s (8.5%; matched-block bootstrap 95% CI, 1.10–2.51 s). The improvement was directionally consistent across operators and objects. Secondary outcomes were interpreted descriptively, and the individual contributions of the two added channels were not isolated. Overall, the proposed framework provides an interpretable one-shot configuration approach for haptic teleoperation of heterogeneous objects.

Keywords: haptic teleoperation, heterogeneous objects, impedance control, object-conditioned configuration, haptic feedback, human-in-the-loop evaluation

1. Introduction

Teleoperation combines human decision-making with robotic execution, enabling remote robots to perform complex manipulation tasks that remain challenging for fully autonomous systems, including hazardous operations, flexible manufacturing, and manipulation in semi-structured or unstructured environments. Haptic feedback conveys remote interaction information to the operator, supporting the perception of contact conditions, grasp stability, and potential object slip. Effective teleoperation therefore requires the coordinated integration of bilateral control, environmental perception, robotic execution, and human–machine interaction [1, 2].

Teleoperation becomes particularly challenging when objects with heterogeneous interaction requirements must be manipulated within the same system. These objects may differ substantially in geometry, deformability, contact sensitivity, and grasp-stability requirements, making a single fixed parameter configuration difficult to reconcile with both compliant contact and stable manipulation. For fragile or deformable objects, lower robot stiffness and requested grasp force may reduce contact impact, but can also compromise positioning accuracy and grasp stability. Conversely, higher stiffness or greater requested grasp force may improve pose retention and grasp stability for rigid or slip-prone objects, but may also increase contact impact or deformation risk. Object categories can therefore serve as compact semantic indicators of task-related interaction requirements and support object-dependent parameter configuration.

Variable-impedance and bilateral-control approaches have been developed to regulate remote-robot behavior according to operator motion, contact state, task phase, and environmental interaction [3–11]. In parallel, visual and multimodal perception have been employed to select task-dependent impedance parameters and generate interaction strategies [12–15], while tactile and kinesthetic feedback have supported operator awareness and grasp execution [16–19]. Collectively, these studies indicate that object- and task-level information can support parameter selection and strategy adaptation.

Despite these advances, many prior approaches have focused on individual aspects of teleoperation, particularly slave-side impedance adaptation or control-authority allocation [4–11, 20–22]. Teleoperation performance, however, depends jointly on remote-robot interaction, the haptic information presented to the operator, and gripper execution. Slave-side impedance regulates contact compliance, master-side haptic feedback shapes the operator's perception of remote interaction, and gripper parameters govern the execution of grasp commands. Although these channels affect different aspects of task execution, they play complementary roles in task efficiency, contact behavior, and

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grasp stability. This observation motivates the use of object-level semantic information to establish an interpretable manipulation strategy and coordinate the initial configuration of multiple task-related channels while retaining operator control.

Direct integration of visual-semantic inference into the high-rate teleoperation loop introduces an additional system-integration challenge. Visual inference generally operates at a lower update rate and may be subject to computational latency and detection fluctuations, whereas the teleoperation loop must remain continuously responsive to operator input. Using each perception result to trigger an immediate parameter update could delay supervisory processing, cause repeated switching, and produce inconsistent interaction behavior. A practical architecture should therefore translate visual-semantic information into a stable task-level strategy without placing visual inference in the high-rate supervisory path.

To address these configuration and integration issues, this paper proposes an **object-conditioned strategy-level configuration framework** for haptic teleoperation of heterogeneous objects. Detected object classes are mapped to manipulation strategies (fragility-priority, balanced, stability-priority), which jointly specify parameter settings for slave-side impedance, master-side haptic feedback, and gripper execution.

Following the first threshold-qualified result read by the supervisory loop, the proposed framework performs a one-shot strategy-level parameter update and retains the selected strategy for the remainder of the trial. This one-shot, strategy-locked design decouples low-rate visual-semantic inference from the nominal 200-Hz supervisory loop. It prevents repeated switching caused by detection fluctuations and maintains a consistent parameter configuration while retaining operator control.

The framework was evaluated on an Omega.7–Franka Panda platform using three operators, six objects, five modes, and 135 trials arranged in 27 matched blocks. Task completion time was the primary endpoint, with the primary comparison between complete multi-channel configuration (mode C) and impedance-only reconfiguration (mode E). Modes A, B, and D provided fixed-parameter, timed manual-selection, and semantic-display reference conditions, respectively.

The main contributions of this work are summarized as follows:

1. **Cross-subsystem strategy-level configuration.** An interpretable strategy layer maps detected object classes to manipulation strategies that jointly configure slave-side impedance, master-side haptic feedback, and gripper execution.
2. **Asynchronous non-blocking strategy deployment.** Bounded queues and non-blocking polling decouple low-rate visual inference from the supervisory loop; one-shot triggering with trial-level strategy locking prevents repeated intra-trial switching.
3. **Matched human-in-the-loop comparative evaluation.** A 27-block, 135-trial, five-mode experiment with three operators and six objects evaluates complete multi-channel configuration against fixed-parameter, manual-selection, semantic-display, and impedance-only reference conditions.

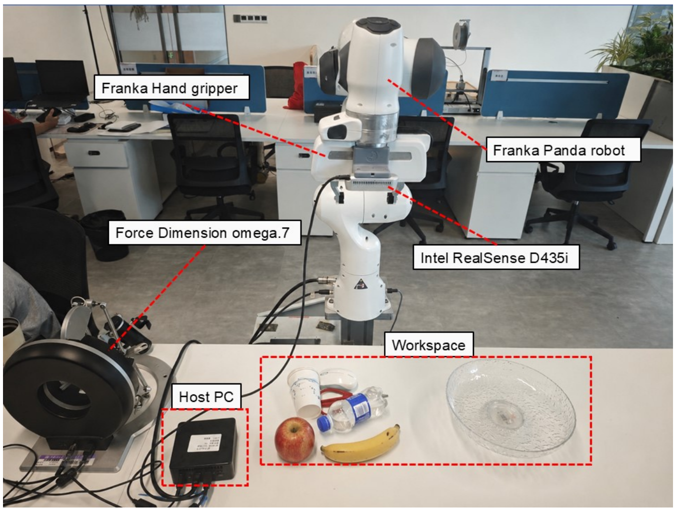

Figure 1: Experimental teleoperation platform showing the Omega.7 master device, a Franka Panda manipulator equipped with a Franka Hand gripper, the Intel RealSense D435i camera, control computer, and object workspace.

The proposed framework targets semi-structured teleoperation scenarios in which the platform and workspace remain fixed while object-level interaction requirements vary. Within this scope, it is intended to support consistent multi-channel configuration and efficient task execution during heterogeneous-object teleoperation.

2. Methods and System Implementation

2.1. System Overview and Multi-Rate Architecture

The experimental platform comprised an Omega.7 force-feedback master device, a seven-degree-of-freedom Franka Panda manipulator equipped with a Franka Hand gripper [23], an Intel RealSense D435i camera, and a control computer (Fig. 1). The D435i acquired 424×240 RGB frames at 15 fps (nominally every 66.7 ms); the depth stream was not used in the present implementation. The control computer executed object detection, the nominal 200-Hz supervisory teleoperation loop, strategy-level parameter configuration, haptic rendering, gripper control, and data logging.

RGB frames were acquired at 15 fps (nominally every 66.7 ms) and passed through a single-slot frame queue to limit stale-frame accumulation. YOLO11n inference ran in a separate process and returned object-class labels and confidence scores through a two-slot result queue. The nominal 200-Hz supervisory loop polled the result queue every 5 ms without blocking for inference, allowing image acquisition, visual inference, and operator-driven teleoperation to proceed concurrently. Per-image processing time was characterized separately using a controlled image test. Table 1 summarizes the operating rates or trigger conditions and associated data flow of the main system modules. Figure 2 illustrates the timing relationships among image acquisition, visual inference, supervisory teleoperation, and strategy-level parameter configuration within the multi-rate asynchronous architecture.

2.2. Teleoperation Mapping and Mechatronic Control Channels

2.2.1. Master–slave motion mapping

Only the translational motion of the Omega.7 was mapped to the Panda end effector. At supervisory step k, the master-device position increment and the desired slave end-effector position were updated as

$$\Delta\mathbf{p}_m(k) = \mathbf{p}_m(k) - \mathbf{p}_m(k-1), \tag{1}$$

$$\mathbf{p}_d(k) = \mathbf{p}_d(k-1) + S\,\mathbf{C}\Delta\mathbf{p}_m(k), \tag{2}$$

where $\mathbf{p}_m(k) \in \mathbb{R}^3$ denotes the measured Omega.7 position, $\Delta\mathbf{p}_m(k) \in \mathbb{R}^3$ is its incremental displacement, and $\mathbf{p}_d(k) \in \mathbb{R}^3$ denotes the desired Panda end-effector position at supervisory step k. The position-scaling factor was $S = 3.0$, and $\mathbf{C} = \mathrm{diag}(-1, -1, 1)$ defined the axis-direction mapping between the master and slave coordinate conventions. The desired end-effector orientation $\mathbf{R}_d$ was held at a prescribed value throughout each trial; operator-commanded rotational motion was not mapped to the Panda.

2.2.2. Slave-side Cartesian impedance control

Following the standard impedance-control formulation [24], a compact task-space representation of the slave-side Cartesian impedance action was written as

$$\mathbf{w}_c = \mathbf{K}(c)\mathbf{e} + \mathbf{D}(c)\dot{\mathbf{e}}, \tag{3}$$

$$\mathbf{K}(c) = \mathrm{diag}(K_t, K_t, K_t, K_r, K_r, K_r), \tag{4}$$

where $\mathbf{w}_c \in \mathbb{R}^6$ denotes the Cartesian impedance action; $\mathbf{e}, \dot{\mathbf{e}} \in \mathbb{R}^6$ are the pose-error and pose-error-rate vectors; and c denotes the detected object class that indexes the active configuration. The matrices $\mathbf{K}(c)$ and $\mathbf{D}(c)$ denote the associated stiffness and effective damping actions, respectively. The scalar values K_t and K_r specify translational and rotational stiffness, respectively. The error vector $\mathbf{e}$ combines translational error relative to the desired position $\mathbf{p}_d$ and rotational error relative to the fixed desired orientation $\mathbf{R}_d$. With $\mathbf{R}_d$ fixed, K_r set the rotational stiffness against disturbances and deviations from the prescribed orientation. No rotational command was supplied by the operator.

The slave-controller API constructed an internal damping quantity $\mathbf{D}_{\mathrm{API}}(c)$ from the active diagonal stiffness matrix $\mathbf{K}(c)$ and the configuration-dependent dimensionless damping-scaling parameter $\zeta(c)$ according to

$$\mathbf{D}_{\mathrm{API}}(c) = 2\zeta(c)\sqrt{\mathbf{K}(c)}, \tag{5}$$

The square root in Eq. (5) was applied elementwise to the diagonal entries of $\mathbf{K}(c)$. $\mathbf{D}_{\mathrm{API}}(c)$ represents the controller-internal damping construction; because no effective Cartesian inertia matrix was identified, $\zeta(c)$ is treated as a dimensionless parameter and is not interpreted as a critical-damping ratio.

2.2.3. Master-side haptic rendering

For master-side haptic rendering, the Franka internal state estimator provided a slave-side external-force estimate; its three translational components $\mathbf{F}_{\mathrm{ext}}$, in newtons, were used as inputs to the force-feedback calculation. The baseline force-feedback command along Cartesian axis i was computed as

$$u_{h,i}^{\mathrm{base}} = \mathrm{sgn}\Big(K_f F_{\mathrm{ext},i}\Big)\max\Big(\big|K_f F_{\mathrm{ext},i}\big| - d, 0\Big), \quad i \in \{x, y, z\}, \tag{6}$$

Here, $u_{h,i}^{\mathrm{base}}$ denotes the baseline master-side force command along axis i, K_f is a dimensionless force-scaling factor, and d is the per-axis dead-zone threshold in newtons. Equation (6) applies force scaling and a symmetric dead zone independently to each Cartesian component.

In addition to the baseline force-feedback command, a separate cue along the positive z-axis was generated from the Omega.7 gripper angle θ_Ω:

$$\alpha_g = \mathrm{clip}\left(\frac{\theta_{\max} - \theta_\Omega}{\theta_{\max} - \theta_{\min}}, 0, 1\right), \tag{7}$$

$$u_g = \min\Big(\alpha_g G_{\mathrm{cue}} F_{\mathrm{cue,max}}, F_{\mathrm{cue,max}}\Big), \qquad u_{h,z} = u_{h,z}^{\mathrm{base}} + u_g. \tag{8}$$

Here, $\alpha_g \in [0, 1]$ is the normalized Omega.7 gripper-aperture variable. With $G_{\mathrm{cue}} = 0.3$ and $F_{\mathrm{cue,max}} = 1.0$ N, $u_g = 0.3\alpha_g$ N, yielding an aperture cue within $[0, 0.3]$ N along the positive z-axis. The cue increased as the Omega.7 gripper moved toward the more-open state and was updated each supervisory cycle. No slave-gripper aperture or independently measured grasp-force signal entered the cue calculation.

2.2.4. Gripper-execution channel

For grasp execution, v_g specified the gripper closing and opening speed, and F_g specified the requested-force input to the Franka Hand `grasp()` API. The target-width argument was fixed at 0 m, and the tolerance parameters `epsilon_inner` (0.005 m) and `epsilon_outer` (0.080 m) remained constant across strategies. After contact and entry into the `HOLDING` state, the gripper maintained the grasp through its internal controller until the prescribed release command. F_g is the requested-force input to the API and does not represent an independently measured fingertip force.

The three parameter channels acted on distinct components of the teleoperation system: slave-side impedance regulated the Panda response to pose error and disturbances, master-side haptic settings mapped the external-force estimate and master-side aperture cue to the Omega.7 force command, and Franka Hand settings specified gripper execution. The proposed framework coordinated these channels through a common strategy-level parameter bundle.

2.3. Object-Conditioned Strategy-Level Parameter Configuration

2.3.1. Object-class-to-strategy and strategy-to-parameter mapping

The framework implemented a two-stage mapping from each detected object class to a manipulation strategy and subsequently

Table 1: Timing, triggers, and data flow of the multi-rate mechatronic system.

Module	Timing or trigger	Data flow
Master input	200 Hz	Omega.7 translational position and button states → master-state update.
Slave control	200 Hz	Desired pose and impedance parameters → Panda command.
Image acquisition	15 fps (66.7 ms/frame)	D435i RGB stream → 424 × 240 RGB frame.
YOLO11n inference	Asynchronous; per-image processing time characterized separately	Latest available frame → object class and confidence score.
Strategy scheduling	First threshold-qualified detection	Object class and confidence → target parameter vector $\mathbf{\Theta}(s)$.
Haptic rendering	200 Hz	Force-related state → Omega.7 force command.
Gripper command	Button event	Gripper input and command parameters → Franka Hand command.

Note: YOLO11n inference ran in a separate process and did not block the nominal 200-Hz supervisory loop.

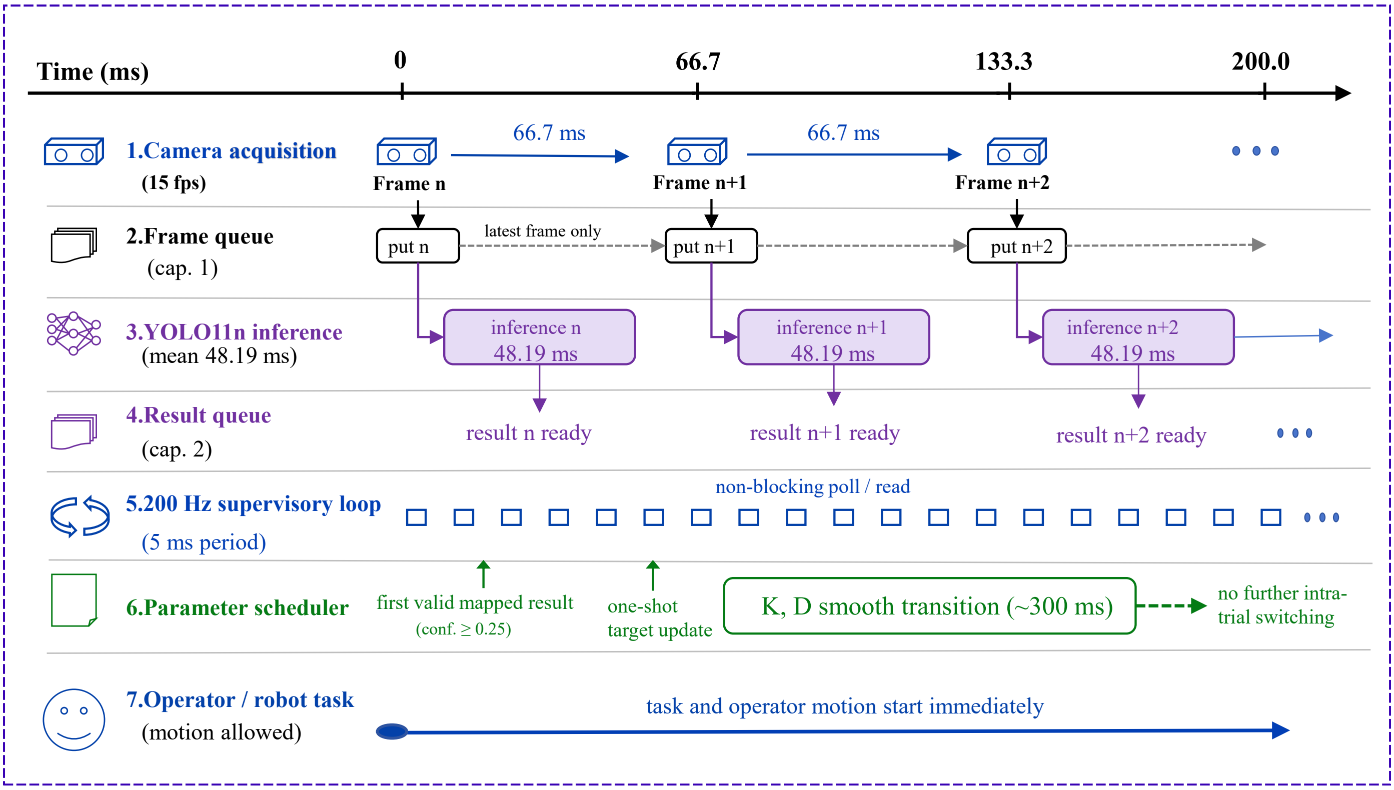

Figure 2: Not-to-scale timing schematic of the asynchronous perception–control architecture. Frames and inference results were exchanged through bounded queues, while the nominal 200-Hz supervisory loop polled the result queue without blocking. The first threshold-qualified result (≥ 0.25 confidence) triggered a single strategy-level update, after which the selected strategy remained locked for the trial; impedance parameters transitioned over approximately 300 ms. The schematic shows nominal module timing; end-to-end trigger latency was not measured.

from the selected strategy to a multi-channel parameter vector:

$$s = f(c), \qquad \mathbf{\Theta}(s) = \begin{bmatrix} K_t(s) \\ K_r(s) \\ \zeta(s) \\ K_f(s) \\ d(s) \\ v_g(s) \\ F_g(s) \end{bmatrix}. \tag{9}$$

Here, c denotes the detected object class, and $s \in \mathcal{S}$ is the corresponding manipulation strategy. The strategy set $\mathcal{S}$ comprised the fragility-priority, balanced, and stability-priority strategies.

The parameter vector $\mathbf{\Theta}(s)$ contains the slave-side impedance parameters $\{K_t, K_r, \zeta\}$, the master-side haptic parameters $\{K_f, d\}$, and the gripper-execution parameters $\{v_g, F_g\}$. For compact notation in the preceding control-channel equations, quantities indexed by c denote values selected through this two-stage mapping.

The three strategies reflected task-relevant interaction priorities—deformation sensitivity, contact impact, slip risk, pose retention, and grasp stability—that motivated the two-object assignment per strategy (Table 2). The assignments were operational labels for the present platform and task, not a general material classification. Parameter-level design rationale is provided in Supplementary Table S1.

2.3.2. One-shot triggering, strategy locking, and parameter transition

At runtime, the first result with confidence ≥ 0.25 retrieved by the supervisory loop triggered a one-shot update of the parameter channels enabled in the active experimental mode. The selected strategy was then locked for the remainder of the trial. Impedance parameters K_t, K_r, and ζ were transitioned over $T_s = 0.30$ s using standard smoothstep interpolation ($h(\tau) = 3\tau^2 - 2\tau^3$):

$$q(t) = q_0 + h(\tau)\left(q_{\text{target}} - q_0\right), \qquad q \in \{K_t, K_r, \zeta\}. \tag{10}$$

The smoothstep interpolation served as command smoothing to avoid abrupt impedance changes [25, 26]. During the transition, $\mathbf{D}_{\text{API}}$ was recomputed from the current values of $\mathbf{K}$ and ζ. The enabled haptic parameters $\{K_f, d\}$ and gripper parameters $\{v_g, F_g\}$ were updated immediately upon the strategy event.

The class-to-strategy mapping was defined manually for the known object classes used in this study. Each recognized class invoked one of the three predefined strategies, providing a bounded and repeatable parameter configuration across the slave robot, haptic interface, and gripper. Online estimation of object physical properties was not included in the mapping. Extending the framework to a new object requires a detector label with reliable recognition and an explicit assignment to one of the predefined strategies before the corresponding parameter vector can be invoked.

Figure 3 summarizes the two-stage object-class-to-strategy-to-parameter mapping, the one-shot strategy event, and the three parameter channels.

2.4. Parameter Sets and Initialization

The initialized baseline parameter vector was

$$\mathbf{\Theta}_0 = [\,150,\ 10,\ 1.0,\ 0.5,\ 0.3,\ 0.10,\ 20\,]^{\mathsf{T}},$$

following the order $[K_t, K_r, \zeta, K_f, d, v_g, F_g]^{\mathsf{T}}$ defined in Eq. (9). $\mathbf{\Theta}_0$ was retained unless replaced by a strategy update; in automatic modes, a threshold-qualified class without an explicit mapping invoked the balanced strategy vector (Table 2).

The strategy-level operating points were selected through engineering screening before formal data collection: candidate settings were excluded when associated with sustained oscillation, noticeable haptic discomfort, grasp failure, or visible object damage. They represent screened operating points for the present platform, objects, and task, not optimized values or damage limits. Parameter-level rationale is provided in Supplementary Table S1.

2.5. Mode-Dependent Runtime Procedure

The supervisory loop executed teleoperation, haptic rendering, gripper control, and data logging across all five modes. Visual inference and the visual-semantic display were enabled only in modes C–E, while mode B used manual strategy selection after trial timing began. Upon a strategy event (automatic in modes C and E, manual in mode B), only the parameter channels enabled by the active mode were updated: mode C updated all three channels, mode E updated only the slave-side impedance channel, and mode B updated all channels manually. Mode D displayed the visual-semantic output while retaining $\mathbf{\Theta}_0$ throughout the trial. In modes B, C, and E, the selected strategy was locked after the first update, teleoperation continued regardless of trial outcome, and the master-side trajectory, gripper input, active parameter state, and task duration were logged. Mode definitions and channel-update logic are summarized in Table 3.

2.6. Fallback Logic and Operational Safeguards

When no threshold-qualified result was available, the initialized parameter state was retained. A result with confidence ≥ 0.25 but without an explicit class-to-strategy mapping invoked the balanced default strategy. Fallback selection was restricted to these predefined targets and only parameter channels enabled by the active mode were updated. Operational safeguards included the robot collision-detection function, zero-force commands on controller exit, a standardized initial pose, and a manually accessible emergency stop. The master-gripper-aperture cue u_g was limited to 0.3 N along the positive z-axis. The baseline force-feedback command was transmitted without an additional application-layer saturation layer.

3. Experimental Design

3.1. Experimental Objectives, Comparisons, and Outcome Hierarchy

The experiment was designed to evaluate the task-level performance of object-conditioned strategy-level parameter configuration during heterogeneous-object haptic teleoperation. Task completion time was specified as the primary endpoint. The primary planned comparison was between the complete multi-channel configuration mode (C) and the impedance-only reconfiguration mode (E).

The primary C–E comparison assessed the system-level contribution of jointly configuring the master-side haptic and gripper-execution channels beyond impedance-only reconfiguration. Because the two added channels changed together, the comparison was interpreted at the bundled system level. Secondary contrasts (C–A, C–B, C–D) provided fixed-operation, manual-selection, and semantic-display references. Observed task success and Raw NASA-TLX were treated as secondary descriptive outcomes; master-side trajectory length and pause count were analyzed as exploratory process measures. All analyses were interpreted as repeated-measures evidence for the three-operator sample.

Table 3 defines the five experimental modes. Mode definitions and channel-update logic are summarized in the table and its note; the C–E comparison was the primary planned contrast.

Table 2: Strategy-level parameter settings and assigned objects. K_t is reported in N/m, K_r in N m/rad, d in N, and v_g in m/s. F_g denotes the requested-force input to `grasp()` in newtons. The damping-scaling parameter ζ and force-scaling factor K_f are dimensionless.

Strategy	Assigned objects	Slave-side impedance	Master-side haptic feedback	Gripper execution
Fragility-priority	Apple, banana	K_t = 50 N/m; K_r = 5 N m/rad; ζ = 0.8	K_f = 0.2; d = 0.3 N	v_g = 0.02 m/s; F_g = 8 N
Balanced	Paper cup, bottle	K_t = 150 N/m; K_r = 10 N m/rad; ζ = 1.0	K_f = 0.5; d = 0.4 N	v_g = 0.05 m/s; F_g = 15 N
Stability-priority	Mouse, scissors	K_t = 200 N/m; K_r = 13 N m/rad; ζ = 1.2	K_f = 0.7; d = 0.5 N	v_g = 0.10 m/s; F_g = 20 N

Note: F_g is the requested-force input to the Franka Hand `grasp()` API and does not represent an independently measured fingertip force.

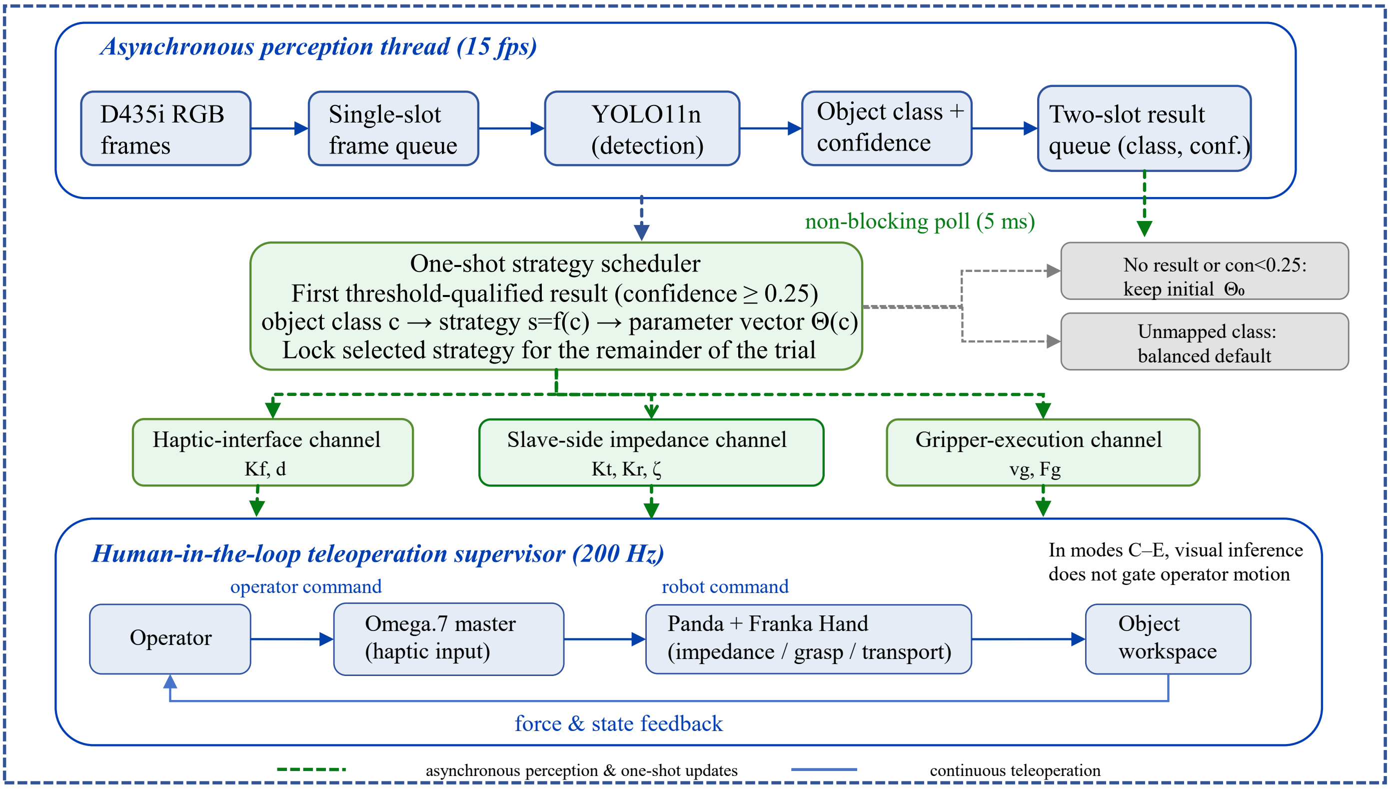

Figure 3: Object-conditioned strategy-level parameter-configuration architecture. D435i RGB frames were processed asynchronously by YOLO11n, and object-class labels and confidence scores were returned through bounded queues. The nominal 200-Hz supervisory loop polled the result queue without blocking. The first result with confidence ≥ 0.25 triggered the object-class-to-strategy mapping and a one-shot update of the enabled slave-side impedance, master-side haptic, and gripper-execution parameter channels. The selected strategy remained locked for the trial. When no threshold-qualified result was available, the initialized parameter state was retained; a detected class without an explicit mapping invoked the balanced default strategy. The continuous operator–robot motion and force-feedback loop remained active throughout task execution.

3.2. Operators, Objects, and Matched-Block Design

Three operators participated in the main experiment (P01–P03; 23–24 years of age; two male and one female; all right-handed). Because the Omega.7 was positioned on the left side of the platform, all operators used their left hand to control the device in every trial. The device position and operating hand were held constant across operators and experimental modes to standardize the physical interface condition. All operators had received basic teleoperation training and completed a 10–15-min familiarization session immediately before formal data collection. During familiarization, they practiced master-device motion, gripper operation, and the complete task procedure. Written informed consent was obtained from all operators before participation.

The six objects were selected to represent different contact, grasping, and transfer requirements, including deformation sensitivity, smooth or irregular surface geometry, and elongated shape. Each object was assigned to the fragility-priority, balanced, or stability-priority strategy to select the corresponding parameter vector for the present platform and task. The assignments served as operational labels for parameter selection in the specified grasp-and-transfer task and were not intended as

Table 3: Experimental modes, information conditions, channel-update logic, and comparison roles.

Mode	Operator information / selection	Slave-side impedance	Master-side haptic feedback	Gripper execution	Comparison role
A: Fixed-Parameter Baseline	No visual-semantic display or strategy selection	Fixed	Fixed	Fixed	Fixed-parameter reference condition.
B: Timed Manual-Selection Workflow	No visual-semantic display; operator identifies the object and selects a complete strategy after timing begins.	Manual	Manual	Manual	Timed manual-workflow reference.
C: Complete Multi-Channel Configuration	Object class, selected strategy, confidence score, and color-coded bounding box	Automatic	Automatic	Automatic	Complete multi-channel condition; primary condition in the C–E comparison.
D: Visual-Semantic Display with Fixed Parameters	Same visual-semantic display as mode C	Fixed	Fixed	Fixed	Semantic-display control condition.
E: Impedance-Only Reconfiguration	Same visual-semantic display as mode C	Automatic	Fixed	Fixed	Impedance-only reference for the primary C–E comparison.

Note: Automatic denotes a one-shot update triggered by the first detection result with confidence ≥ 0.25. Manual denotes a one-shot complete-strategy update after operator selection, with identification and selection included in trial timing. Fixed denotes retention of the initialized parameter state. In modes C–E, no result or confidence below 0.25 produced no parameter update. A threshold-qualified class without an explicit mapping invoked the balanced default strategy; mode D displayed the visual-semantic output while retaining $\mathbf{\Theta}_0$. Visual inference was not used in modes A or B. The C–E comparison was interpreted at the bundled system level because the master-side haptic and gripper-execution channels changed together.

a general classification of material properties. Table 4 summarizes the approximate physical profiles, assigned strategies, and principal manipulation considerations for the six objects. The paper cup and scissors provide contrasting examples of deformation-sensitive handling and orientation-sensitive transfer, respectively.

The main experiment comprised 27 matched task blocks. Each block was defined by a unique combination of operator, object, and repetition index and contained one trial under each of modes A–E. Accordingly, 27 blocks × 5 modes per block yielded 135 trials. The five mode-specific observations within each block therefore formed the matched set for the completion-time analysis.

For each operator and strategy category, three matched blocks were allocated between the two assigned objects in a 2:1 ratio, reversed across operators (Table 5). The same object was used for all five mode trials within a block, preserving object-matched comparisons.

Mode orders were preassigned using nine different sequences across the operator × strategy groups and were not fully counterbalanced (Supplementary Table S2). With three operators, mode order, object allocation, and operator effects could not be estimated independently; the C–E comparison was therefore treated as a paired contrast within the observed sample.

3.3. Task Procedure and Outcome Measures

Before each trial, the test object was manually placed within a predefined workspace region and aligned to the prescribed orientation as closely as practicable. Small trial-to-trial variations in object position and orientation remained after manual placement; the same placement protocol was applied across all five modes. Only translational master-device motion was mapped to the Panda end effector, while the desired end-effector orientation remained fixed throughout the trial.

Each trial comprised six stages: reset, approach, grasp, transfer, release, and termination. In modes C–E, camera acquisition, visual inference, and teleoperation began concurrently at task initiation. Operator-driven motion was permitted while visual inference proceeded asynchronously. In mode B, trial timing began before manual object identification and button-based strategy selection. Robot motion was enabled after the operator completed the selection step. The end of the manual selection step was not recorded with a separate timestamp.

A trial was classified as successful when the object was grasped, transferred, and placed without a drop, observable slip, or visible damage. An occurrence of a drop, slip, or damage was recorded as a failure. Failure labels were assigned by a designated researcher from direct observation; systematic video recording and independent blinded re-evaluation were not performed (see Section 5.5). Failures did not terminate trial timing early, regrasping was not permitted, and successful and failed trials shared the same predefined timing endpoint. The retained dataset included the master-side trajectory, gripper input, active parameter state, task duration, and outcome label. Event-level object-class labels, confidence scores, strategy-trigger timestamps, and synchronized trigger-to-contact timestamps were not logged (Section 5.5).

Task completion time was measured from task initiation to the common predefined terminal state and was retained for both successful and failed trials. Observed success rate was calculated as the number of successful trials divided by the total number of trials in each mode. It was reported descriptively because the outcome labels were assigned from direct observation without independent re-evaluation.

Subjective workload was assessed using the Raw NASA-TLX questionnaire [27]. Each of the six dimensions was rated on a 0–100 scale, and the unweighted arithmetic mean of the six ratings was used as the Raw NASA-TLX score. Each operator completed one questionnaire immediately after completing the three trials associated with a given strategy × mode condition. The three trials combined the two objects assigned to that strategy according to the operator-specific 2:1 allocation. The workload dataset therefore comprised 3 operators ×3 strategies ×5 modes = 45 questionnaires, corresponding to nine questionnaire units per mode. This questionnaire structure supported descriptive comparisons at the mode and strategy levels. Object-

Table 4: Test objects, assigned strategies, approximate physical profiles, and principal manipulation considerations.

Object	Assigned strategy	Approximate physical profile	Principal manipulation consideration
Apple	Fragility-priority	~200 g; smooth surface; diameter 70–80 mm	Contact impact and slip; gentle grasping required.
Banana	Fragility-priority	~120 g; smooth surface; 20×160 mm	Compression-related deformation and slip.
Paper cup	Balanced	~5 g; paper; diameter 55 mm; height 60 mm	High deformation sensitivity with a concurrent stability requirement.
Bottle	Balanced	~30 g; smooth plastic; diameter 65 mm; height 100 mm	Transfer slip and the balance between gentle contact and grasp stability.
Mouse	Stability-priority	~100 g; smooth plastic; $65 \times 100 \times 35$ mm	Rigid, irregular geometry and potential transfer slip.
Scissors	Stability-priority	~150 g; metal and plastic; $50 \times 130 \times 15$ mm	Elongated rigid geometry and a high orientation-retention requirement.

Table 5: Distribution of matched task blocks and trials across strategies and objects.

Object	Blocks	Trials
Fragility-priority		
Apple	4	20
Banana	5	25
Balanced		
Paper cup	5	25
Bottle	4	20
Stability-priority		
Mouse	5	25
Scissors	4	20
Total (six objects)	27	135

Note: Each matched block contains five trials—one under each experimental mode—so the numbers of trials equal five times the corresponding numbers of blocks.

specific workload was not measured separately.

Master-side trajectory length and pause count were analyzed as exploratory process measures. Visual performance was characterized separately in a controlled image test using object-class accuracy, class-to-strategy mapping accuracy, confidence score, and per-image processing time. Before analysis, a pause was operationally defined as a continuous interval during which the master-device translational speed remained below 0.005 m/s for at least 0.30 s. Master-device speed was obtained by differentiating the raw master-side position trajectories, which were sampled at approximately 200 Hz; pause intervals were then identified using the predefined speed and duration thresholds.

3.4. Statistical Analysis

Completion time across the five modes was analyzed using a Friedman test, treating the 27 operator-nested matched task blocks as the repeated-measures units. When the omnibus Friedman test was significant, the four planned paired contrasts comparing mode C with modes A, B, D, and E were evaluated using Wilcoxon signed-rank tests with Holm adjustment for multiple comparisons.

For each matched block b, the primary C–E completion-time difference was defined as

$$\Delta T_b = T_{E,b} - T_{C,b}, \tag{11}$$

so that a positive value favored mode C. The analysis reported the mean paired difference $\overline{\Delta T}$, the relative reduction with mode E as the reference, and a matched-block percentile bootstrap 95% confidence interval. The interval was calculated from 10,000 resamples with replacement of the 27 matched-block differences. Operator-specific mean paired differences were reported to assess directional consistency across operators. A leave-one-operator-out sensitivity analysis was additionally performed to evaluate whether the overall C–E difference was dominated by any single operator.

Observed success outcomes were summarized as counts and percentages for each mode, and paired C–E outcomes were cross-tabulated within the matched blocks. Given the small number of discordant C–E pairs, the exact two-sided McNemar test was treated as a supplementary analysis. Raw NASA-TLX was summarized descriptively across the nine operator × strategy questionnaire units per mode; no inferential test or bootstrap confidence interval was calculated.

Master-side trajectory length and pause count were treated as exploratory process measures. For the C–E trajectory-length comparison, the mean paired difference and a matched-block percentile bootstrap 95% confidence interval were reported using the same 10,000-resample procedure. Pause counts were summarized descriptively; no formal hypothesis test was prespecified or performed. Non-binary outcomes were summarized using median [interquartile range (IQR)] and mean ± standard deviation (SD), as appropriate; categorical success outcomes were reported as counts and percentages. Because the 27 matched blocks were nested within only three operators, the resulting p-values and matched-block bootstrap intervals were interpreted conditionally for the observed repeated-measures sample and were not treated as estimates of population-level participant effects.

4. Results

4.1. Controlled Visual Test and Supervisory-Loop Timing

The separate controlled closed-set visual test used 180 photographs, with 30 images per object class, acquired under fixed viewpoint, background, and lighting conditions. All 180 images were classified correctly and mapped to the intended manipulation strategy. The mean confidence score was 0.853, and the mean per-image processing time was 48.19 ms (Fig. 4). Class-wise results and the confusion matrix are provided in Supplementary Table S3 and Supplementary Fig. S1.

These results characterize closed-set recognition and strategy mapping for the six tested object instances under the specified controlled imaging conditions. They do not quantify the correctness of online strategy triggering, trigger-to-contact timing, or performance on untested objects.

The median logged duration of the nominal 200-Hz supervisory cycle was 5.07 ms.

4.2. Overall Results Across the Five Modes

Across the five modes, mode C showed the lowest median completion time, the highest observed success rate, the lowest median Raw NASA-TLX score, and the lowest median master-side trajectory length (Table 6 and Figs. 5–6). Using the trial-level data in Supplementary Table S4, the mean completion time was 19.28 ± 1.30 s in mode C, compared with 21.42 ± 1.58 s in mode A, 21.01 ± 1.61 s in mode B, 20.91 ± 1.10 s in mode D, and 21.07 ± 1.56 s in mode E. With each comparison mode used as the reference, the corresponding mean reductions for mode C were 10.0%, 8.2%, 7.8%, and 8.5%, respectively. Consistent with the predefined outcome hierarchy, inferential analyses focused on task completion time as the primary endpoint.

Using the 27 operator-nested matched task blocks as repeated-measures units, the Friedman test indicated an overall difference in completion time across the five modes ($\chi^2(4)$ = 30.904, $p < 0.001$). Holm-adjusted Wilcoxon signed-rank tests indicated lower completion times in mode C than in modes A, B, D, and E, with all adjusted p-values below 0.01. These comparisons were interpreted as conditional repeated-measures evidence for the observed three-operator sample. Descriptively, mode C yielded the lowest Raw NASA-TLX score within each of the nine operator × strategy questionnaire units.

4.3. Primary C–E Comparison and Exploratory Process Measures

The primary paired comparison between modes C and E is summarized in Table 7 and Fig. 7.

Median task completion time was 19.57 s in mode C and 20.73 s in mode E. Defining the block-level difference as $\Delta T = T_E - T_C$, the mean paired difference was 1.79 s, with a matched-block bootstrap 95% confidence interval of 1.10–2.51 s. Relative to the mean completion time in mode E, this difference corresponded to an 8.5% reduction in mode C. Mode C had a lower completion time than mode E in 22 of the 27 matched task blocks.

A successful-trial sensitivity analysis (23 blocks with both trials successful) yielded a similar estimate (mean paired difference 1.80 s; 19/23 blocks favoring mode C).

For the paired success outcomes, 23 matched blocks were successful in both modes, three were successful in mode C only, one was successful in mode E only, and none failed in both modes. The exact two-sided McNemar test yielded p = 0.625 and, given only four discordant pairs, did not provide clear evidence of a mode-related difference in observed success. Accordingly, the observed success-rate difference was treated as supplementary descriptive evidence.

Median master-side trajectory length was 0.697 m in mode C and 0.732 m in mode E. With $\Delta L = L_E - L_C$, the mean paired trajectory-length difference was 0.024 m, with a matched-block bootstrap 95% confidence interval of -0.014 to 0.059 m. Because the interval included zero, the data did not provide clear evidence of a C–E difference in master-side trajectory length. Pause counts were lower on average in mode C, although the two modes had the same median pause count.

Using the predefined pause criterion in Section 3.3, mode C had a median [IQR] pause count of 3 [2, 3.5] and a mean of 2.74 ± 1.23, whereas mode E had a median pause count of 3 and a mean of 3.41 ± 1.67. The mean pause count was lower in mode C within each of the three manipulation strategies, while the overall median remained 3 in both modes. Pause count was treated descriptively, and no formal hypothesis test was performed. The robustness of the result to alternative speed or duration thresholds was not evaluated.

4.4. Observed Failure Outcomes

Across the 135 trials, 18 were classified as failures: 5, 6, 1, 3, and 3 in modes A–E, respectively. The observed failure outcomes involved object drops, observable slip, or visible damage according to the predefined criteria in Section 3.3. Table 8 summarizes the mode-level failure counts and representative trial observations.

Among the five modes, mode C had the lowest observed failure count, with 1 failure in 27 trials.

4.5. Directional Consistency Across Operators and Objects

The mean paired C–E completion-time difference favored mode C for each of the three operators and each of the six tested objects.

At the operator level, the mean paired differences $\Delta T = T_E - T_C$ were 1.66, 2.56, and 1.16 s for P01, P02, and P03, respectively (mode C faster in 7/9, 9/9, and 6/9 matched blocks). Leave-one-operator-out mean paired differences were 1.86, 1.41, and 2.11 s, all remaining positive. Operator-level medians [IQR] are shown in Fig. 8.

At the object level, mode C had the lowest mean task completion time among the five modes for each of the six tested objects (Supplementary Table S5). Relative to mode E, the mean completion-time reduction in mode C ranged from 3.3% for the bottle to 13.2% for the scissors. The mean C–E difference favored mode C for all six objects, while the magnitude varied across objects.

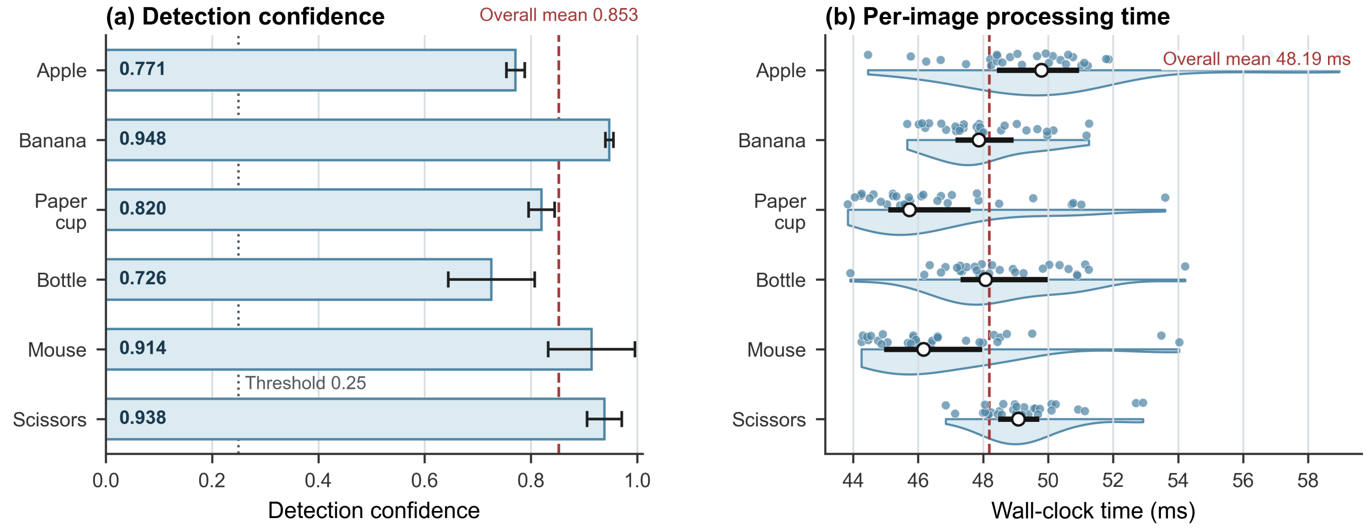

Figure 4: Confidence scores and per-image processing time in the controlled closed-set visual test (n = 30 images per object class). Bars indicate mean ± SD; half violins, points, segments, and open circles indicate distributions, individual observations, IQRs, and medians, respectively. Reference lines indicate the confidence threshold of 0.25 and the overall means for confidence (0.853) and processing time (48.19 ms). The reported processing time does not represent end-to-end task-trigger latency.

Table 6: Descriptive task-performance and workload results across the five experimental modes.

Mode	Task completion time (s)	Master-side trajectory length (m)	Observed success, n/N (%)	Raw NASA-TLX
A	21.18 [20.62, 22.08]	0.757 [0.693, 0.816]	22/27 (81.5%)	62.50 [59.67, 64.50]
B	20.89 [20.12, 21.83]	0.787 [0.721, 0.861]	21/27 (77.8%)	56.17 [55.00, 59.33]
C	**19.57 [18.41, 20.05]**	**0.697 [0.660, 0.769]**	**26/27 (96.3%)**	**48.67 [47.67, 51.83]**
D	20.79 [20.32, 21.16]	0.722 [0.678, 0.768]	24/27 (88.9%)	60.33 [57.33, 62.50]
E	20.73 [19.95, 22.25]	0.732 [0.678, 0.799]	24/27 (88.9%)	53.67 [51.83, 57.83]

Note: Continuous outcomes are reported as median [IQR], whereas observed success is reported as successful trials/total trials (%). Task completion time, master-side trajectory length, and observed success use n = 27 trial-level observations per mode. Raw NASA-TLX uses n = 9 operator × strategy questionnaire units per mode. Mode definitions are provided in Table 3, and the trial-level task-performance data are provided in Supplementary Table S4.

Table 7: Primary within-sample comparison of complete multi-channel configuration and impedance-only reconfiguration.

Outcome	Mode C, median [IQR]	Mode E, median [IQR]	Mean paired difference Δ (E–C) [95% CI]	Operator-level direction
Task completion time (s)	19.57 [18.41, 20.05]	20.73 [19.95, 22.25]	1.79 [1.10, 2.51]	3/3 favored C
Master-side trajectory length (m)	0.697 [0.660, 0.769]	0.732 [0.678, 0.799]	0.024 [-0.014, 0.059]	Mixed
Raw NASA-TLX	48.67 [47.67, 51.83]	53.67 [51.83, 57.83]	4.87 [—]	3/3 favored C

Note: Differences were defined as mode E minus mode C, so positive Δ values favor mode C for all three outcomes. Task completion time and master-side trajectory length use n = 27 matched task blocks. Raw NASA-TLX uses n = 9 paired operator × strategy questionnaire units per mode. Confidence intervals for completion time and trajectory length are matched-block bootstrap 95% intervals based on 10,000 resamples and are conditional on the observed block structure. No bootstrap confidence interval was calculated for Raw NASA-TLX.

5. Discussion

5.1. Principal Findings and Interpretation of the Five-Mode Design

In the primary C–E comparison, complete multi-channel configuration reduced mean completion time by 1.79 s (8.5%; bootstrap 95% CI 1.10–2.51 s), and this direction was consistent across all operators and objects. The five-mode design isolates the contribution of parameter configuration at three levels: C–D separates semantic display from parameter configuration, C–E bundles the master-side haptic and gripper-execution channels with impedance, and C–A/C–B provide fixed-operation and manual-selection references. Because the two added channels changed together, the C–E comparison is interpreted at the bundled system level.

Table 8: Observed failure counts and representative trial observations by experimental mode.

Mode	Observed failures, n/N	Representative trial observations
A	5/27	Paper-cup deformation and difficulty maintaining the scissors' pose were noted in failed trials.
B	6/27	No consistent failure pattern could be assigned from the retained trial records.
C	**1/27**	Observable mouse slip during transfer.
D	3/27	Object drops or loss of grasp after grasp establishment.
E	3/27	Observed drops, slip, or visible damage during grasp establishment or transfer.

Note: Mode definitions are provided in Table 3. Failure labels were assigned during the experiment using the predefined criteria in Section 3.3. The listed observations are representative rather than exhaustive. Systematic video recording and independent blinded re-evaluation were not performed; the observations should not be interpreted as confirmed root-cause classifications.

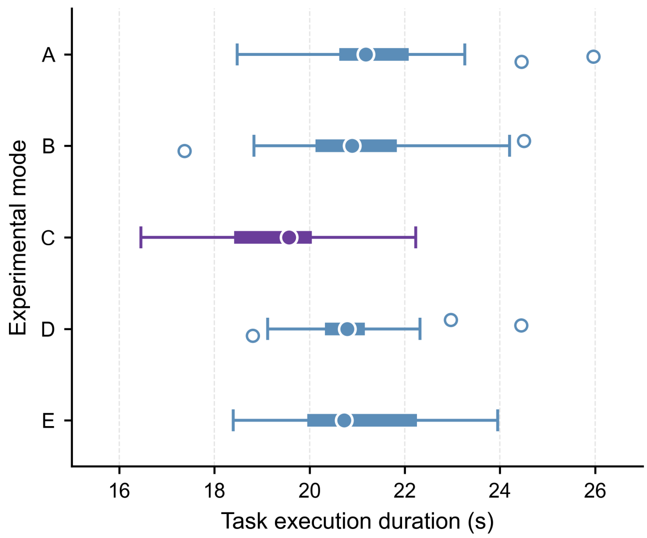

Figure 5: Task completion time across the five experimental modes (n = 27 matched task blocks per mode). Filled circles, thick horizontal segments, whiskers, and open circles indicate medians, IQRs, 1.5×IQR ranges, and outliers, respectively. This descriptive panel displays marginal mode distributions; the paired C–E observations are shown in Fig. 7.

The trajectory-length confidence interval included zero, and the two modes had the same median pause count (although the mean was lower in mode C). This pattern suggests that the time reduction may be associated with fewer corrective interruptions in some trials rather than shorter geometric paths; this process-level interpretation remains exploratory.

5.2. Asynchronous Perception–Control Integration

Because visual output was used only for the one-shot strategy event, the perception process did not need to operate at the supervisory rate. The multi-rate architecture decoupled 15-fps visual inference from the 200-Hz supervisory loop through bounded queues and non-blocking polling, placing semantic perception outside the high-rate critical path. The mean per-image processing time (48.19 ms) substantially exceeded one supervisory period (median 5.07 ms), highlighting the role of this decoupling. The architecture was designed for one-shot strategy configuration with subsequent locking; continuous visual servoing was outside the implemented control structure. Hard real-time guarantees and end-to-end trigger-to-contact timing remain uncharacterized.

5.3. Variation Across Operators and Objects

The positive mean C–E difference was consistent across all three operators and persisted in leave-one-operator-out analyses (Fig. 8). Mode C also had a lower mean completion time than mode E for each of the six objects, with descriptive reductions ranging from 3.3% (bottle) to 13.2% (scissors). The larger reductions for the paper cup and scissors are compatible with greater value from complete multi-channel configuration when deformation management or orientation retention becomes more demanding. Each object-level summary was based on four or five matched blocks and is therefore interpreted descriptively.

5.4. Relation to Previous Work and Mechatronic Design Implications

Prior tele-impedance and variable-impedance studies have adapted slave-side impedance using human-state information, contact-force information, demonstrations, or task phase [4–11]; vision-based methods have mapped object attributes, geometry, or vision–language representations to stiffness or impedance settings [12–15]. Human-commanded stiffness interfaces can also exhibit coupling with force feedback in bilateral tele-impedance [28]. Related work on automatic switching, shared control, multi-stiffness interfaces, and teleoperated grasping has addressed mode selection, control-authority allocation, and gripper impedance based on tactile deformation or grip-safety information [19–22, 29, 30].

The present framework complements these directions by using a single object-conditioned strategy event to coordinate a strategy-level operating state across slave-side impedance, master-side haptic feedback, and gripper execution. The principal contribution is therefore a cross-subsystem, strategy-level configuration architecture together with a system-level evaluation of the resulting multi-channel coordination. At the software-architecture level, bounded queues, non-blocking polling, one-shot triggering, and strategy locking decouple low-rate semantic perception from the nominal 200-Hz supervisory execution path and prevent repeated parameter switching within a trial.

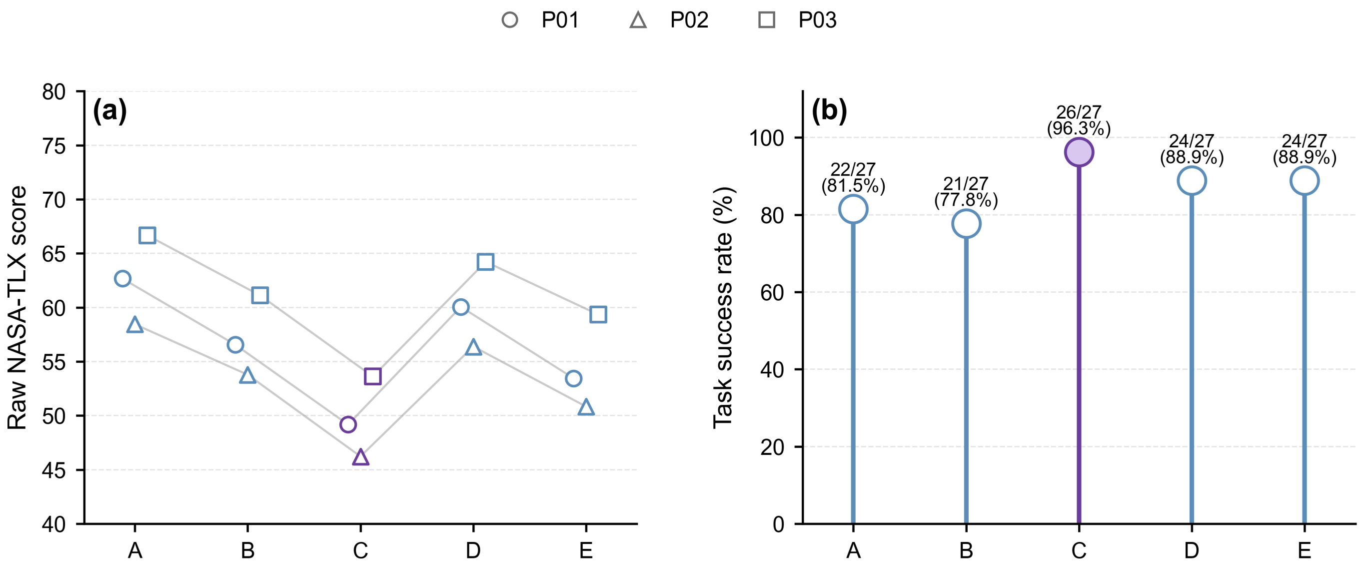

Figure 6: Descriptive workload and observed task-success results across the five experimental modes. (a) Raw NASA-TLX scores: connected markers show each operator's mean across the three strategy-level questionnaire units within each mode (n = 9 operator × strategy units per mode), and marker shape identifies the operator. (b) Number of successful trials out of 27 trials per mode. Both panels are descriptive; no confidence intervals or primary inferential tests are shown.

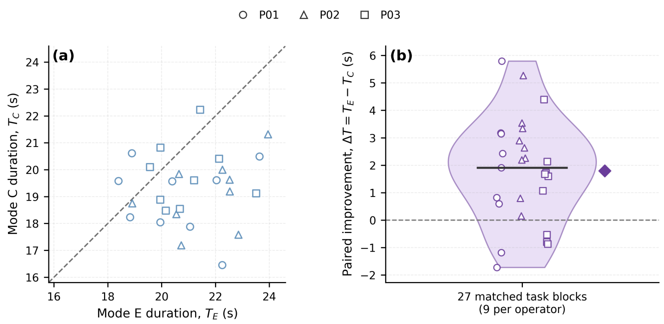

Figure 7: Paired task completion times for modes C and E across 27 matched task blocks. Points below the identity line indicate shorter completion time in mode C. For the paired difference $\Delta T = T_E - T_C$, positive values favor mode C. The violin, horizontal segment, and diamond show the distribution, median, and mean of the paired differences, respectively. The mean paired difference was 1.79 s (matched-block bootstrap 95% CI, 1.10–2.51 s), and 22 of 27 blocks favored mode C.

This design principle may be useful in semi-structured applications such as flexible sorting, remote maintenance, and dismantling, where a limited set of recurring object or task classes can be associated with predefined and experimentally screened operating strategies. Extending the framework to larger class sets would require systematic maintenance of the object-class-to-strategy mapping and may require a richer strategy space than the three levels evaluated here. Tactile or contact-state sensing could be added as a post-contact verification layer, enabling corrective updates after contact without requiring the visual process to operate within the nominal 200-Hz supervisory execution path.

5.5. *Scope and Future Extensions*

The experimental evidence was obtained from three operators, six known objects, translational master motion, a fixed end-effector orientation, prescribed object placement, and pre-assigned mode orders without complete counterbalancing. The class-to-strategy mapping was manually defined. The findings therefore characterize the present platform and task within the tested sample.

Several measurement and design limitations should be considered in future work. First, because the C–E contrast changed the master-side haptic and gripper-execution channels together, separate channel-level ablations are required to resolve their individual contributions. Second, the trial logs did not retain event-level object-class labels, confidence scores, or synchronized strategy-event/contact timestamps; trial-level verification of online trigger timing requires synchronized visual, strategy-event, and contact logging. Third, outcome labels were based on direct observation without independent video-based re-rating. Fourth, although smoothstep interpolation was used for impedance transitions, no formal transient-stability analysis of the complete teleoperation system was performed. Finally, the retained haptic data characterize the commanded signal but not realized device output, and no saturation-status signal was recorded. Future experiments should combine synchronized logging, independent video-based outcome assessment, formal stability analysis, direct contact and realized haptic-output measurements, saturation monitoring, and systematic parameter optimization. An evidence-boundary overview is provided in Supplementary Table S6.

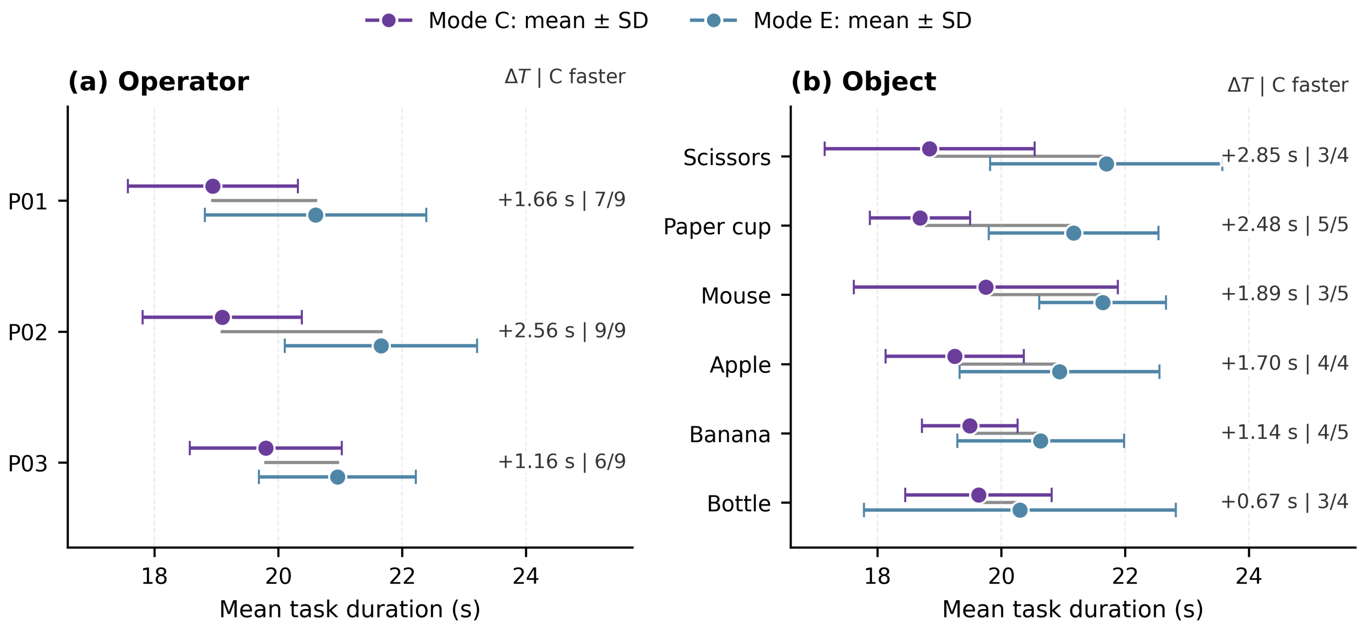

Figure 8: Operator- and object-level summaries of task completion time for modes C and E. (a) Mean task completion time by operator (n = 9 matched task blocks per operator). (b) Mean task completion time by object (n = 4 or 5 matched task blocks per object). Markers and horizontal error bars indicate mean ± SD for each mode. Right-side annotations report the mean paired difference ($\Delta T = T_E - T_C$) and the number of blocks in which mode C was faster divided by the total number of blocks. Positive ΔT values favor mode C. Error bars show within-group variability and are not confidence intervals for the paired differences.

6. Conclusion

This study presented an object-conditioned strategy-level parameter-configuration framework that coordinates slave-side impedance, master-side haptic feedback, and gripper execution through a one-shot, strategy-locked deployment from asynchronous visual inference, using an intermediate object-to-strategy mapping as the coordinating mechanism.

Across 27 matched blocks (135 trials), complete multi-channel configuration reduced mean task completion time by 1.79 s (8.5%; bootstrap 95% CI 1.10–2.51 s) compared with impedance-only reconfiguration, with directional consistency across all three operators and all six tested objects. Secondary descriptive outcomes (success rate and Raw NASA-TLX) also favored the complete configuration, while the trajectory-length confidence interval included zero.

These findings support a within-sample, bundled system-level benefit of coordinating the three parameter channels under the tested conditions. Individual channel contributions remain to be isolated through ablation experiments.

Supplementary Material

Supplementary Tables S1–S6 and Supplementary Fig. S1 are provided in a separate supplementary file.

Declarations

Ethics statement

The school confirmed that formal ethics committee review was not required for this low-risk, non-invasive study. All participants provided written informed consent before participation. The experimental procedure was conducted in accordance with relevant institutional guidelines; no personally identifiable information was collected, and all data were anonymized.

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The author(s) used ChatGPT for grammar and language checking only. No scientific content was generated using AI. The author(s) reviewed and edited the content and take full responsibility for the published article.

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